MALTA MICROSOFT AI USER GROUP

MMAUG Secure Login Portal — Technical Proposal and Deployment Record

Response: MMAUG-AIDB-RFP-002
Version: 5.2 — Current working implementation
Date: 14 November 2026

> **Educational-use caveat:** The MMAUG Secure Login Portal is a solution built as part of the Malta Microsoft AI User Group (MMUAG) AI & DevOps Fundamentals Bootcamp 2026. The solution is for demo/educational purposes and not to be adopted as a production-ready system.

Executive proposal

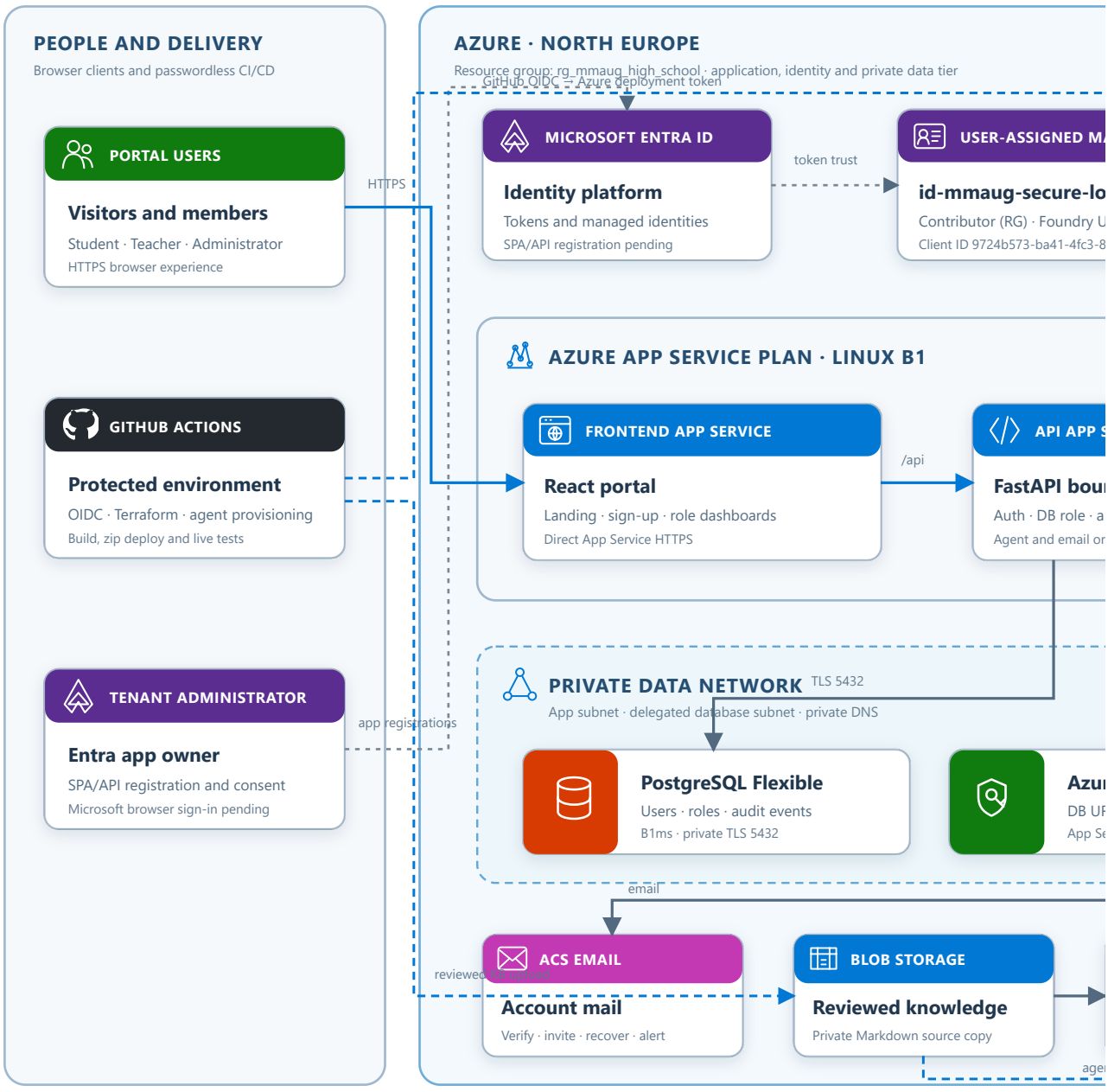
The solution provides a public school-service landing page and protected Student, Teacher and Administrator workspaces. React supplies the frontend, FastAPI enforces the security boundary, and private PostgreSQL stores user status, identity mappings, roles, audit events, rate limits and local credentials.

The development environment is live at `https://app-mmaug-development-web-gkt6d.azurewebsites.net`. Local email accounts, verification, recovery, invitations, administrator approval, user deletion, per-user activity and role dashboards are deployed. Microsoft sign-in remains disabled until the tenant administrator supplies the SPA/API registration client IDs.

GitHub Actions deploys through OIDC to the single approved managed identity. The same identity is attached to the API for approved Azure service access. No Azure deployment secret or App Service publishing credential is stored in GitHub.

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Current working Microsoft Azure architecture · 14 November 2026



Authorization and trust boundary

FastAPI validates identity and account state, reads the active role from PostgreSQL, applies guardrails, and selects one fixed Foundry agent. The browser and mode

Current Azure architecture

Area	Deployed state
Azure scope	Tenant `71beefd7-29f4-4b9e-9cb2-e78b5dc851ad`, `rg_mmaug_high_school`; app/data in North Europe and Foundry in Sweden Central
Public URL	Direct frontend App Service HTTPS URL
Hosting	Linux B1 App Service Plan; separate React and FastAPI Web Apps
Deployment identity	`id-mmaug-secure-login-portal`; GitHub environment OIDC; Contributor at resource-group scope

Area	Deployed state
Database	PostgreSQL Flexible Server 16, B1ms, VNet injected, private DNS, public access disabled
Secrets	Key Vault references for database and local-token values
Local identity	Argon2id password hashes, verified email, short-lived versioned JWTs
Email	Azure Communication Services using managed identity and Azure-managed sender
Monitoring	Workspace-based Application Insights and Log Analytics
AI platform	Sweden Central Foundry project, `gpt-5.4-mini` DataZoneStandard capacity 10 and four running prompt agents
AI knowledge	Reviewed Markdown in private Blob storage and four audience-isolated Foundry File Search stores
Current AI behavior	Live visitor/student/teacher/admin agents, server-selected role, deterministic guardrails and constrained outage fallback

Identity, enrollment and roles

Authentication and authorization remain separate. A valid local or future Microsoft identity does not grant portal access. FastAPI reads the database record for every protected request and rejects inactive or unauthorized users.

- New self-service accounts start inactive as Student and cannot request another role.
- Local users verify mailbox ownership before activation or role change.
- Administrators invite or re-invite people, select a database role, activate or deactivate access, delete other users and review audit activity.
- Students and teachers can view only activity performed by or targeting their own account.
- An administrator cannot delete or deactivate their own active portal account.
- Microsoft identities will be keyed by immutable tenant ID and object ID after MSAL is activated.

The browser never selects or asserts a role. PostgreSQL is the authorization source of truth, and direct SQL role changes are excluded from normal operation because they bypass verification and audit controls.

Account communications

Azure Communication Services sends role invitations, mailbox verification, password setup, recovery and administrator verification alerts. The API authenticates with the managed identity and waits for the final ACS operation result before returning a success message. Re-sending rotates unused setup or verification links.

This remains a demonstration email configuration. A verified school-owned sending domain, delivery/suppression monitoring and operating procedures are prerequisites for broader use.

Assistant design and guardrails

The landing page exposes a Visitor assistant, and signed-in users receive Student, Teacher or Administrator guidance selected by FastAPI from the PostgreSQL role. The client cannot choose an agent identity.

Reviewed guidance is separated by audience. Before any model call, deterministic controls reject prompt injection, secrets, cross-user data and role-changing actions. Assistants have no database connection or state-changing tools, and question text is not written to audit details.

The active Foundry account/project is in Sweden Central. The approved API identity has effective **Foundry User**; all four agents passed direct smoke tests and the deployed public API returned `mode: "foundry"`. Each agent uses a separate File Search vector store. The fallback remains as an availability control, not as the normal execution path.

Delivery model and separation of duties

The `development` GitHub environment federates to the approved managed identity. The workflow validates Terraform, publishes reviewed knowledge, packages both applications, deploys them and verifies live health. CI separately runs backend tests, coverage, lint, frontend build and security checks.

The tenant administrator retains ownership of:

- the SPA and API app registrations and Enterprise Applications;
- delegated `access_as_user` scope and consent;
- local and deployed SPA redirect URIs;
- confirmation of supported Microsoft account types; and
- Azure RBAC assignments that Contributor cannot grant.

The SPA and API registrations require client IDs, not client secrets. Client IDs belong in protected GitHub environment variables. The managed identity is deliberately not given Microsoft Graph directory permissions.

Network and data protection

PostgreSQL has no public endpoint. The API reaches it through App Service VNet integration, the delegated database subnet and linked private DNS. Authorized database inspection uses the API App Service SSH console or another approved client with private-network routing.

Database and JWT values are generated or stored in Key Vault and referenced by the API. Passwords are Argon2id hashes; one-time values are stored as SHA-256 digests. Persistent rate limits protect registration, sign-in, resend, verification and recovery endpoints.

Application Insights and Log Analytics collect service health, requests, dependencies and exceptions. Portal audit records cover application actions; Azure telemetry is not used as the portal authorization source.

Current public ingress

The working public entry point is the frontend Azure App Service HTTPS URL. The current architecture and acceptance record describe that deployed path only.

Acceptance record

Verified

1. GitHub OIDC authenticates as the approved single managed identity.
2. Terraform and application deployment complete through the protected workflow.
3. Frontend and PostgreSQL-backed API health return HTTP 200.
4. Local registration, verification, recovery, invitations and database role approval are implemented with rate limiting and audit events.
5. Student, Teacher and Administrator dashboards and role-aware guardrails are deployed.
6. PostgreSQL is private and the API uses VNet integration.
7. ACS accepts managed-identity email operations; the application now tracks their final operation result.
8. Sweden Central Foundry project, `gpt-5.4-mini` model and four role agents are provisioned.
9. The four direct Foundry smoke tests passed with isolated File Search knowledge.
10. The live App Service public assistant returned `mode: "foundry"` on 14 November 2026.

Remaining gates

1. Tenant administrator completes the SPA/API registrations and supplies both client IDs.
2. Connect Foundry tracing to Application Insights and complete alert/cost thresholds.
3. Load, accessibility, backup/restore, privacy, retention, penetration, Conditional Access, incident-response and operational-ownership reviews are completed.

Real student data remains prohibited until those governance and assurance gates are formally approved.